AMARNATH SAH

Residence: Kattankulathur, Chennai, Tamil Nadu, India
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Education

B.Tech in Computer Science and Engineering — *SRM Institute of Science and Technology*
Computer Science Engineering (AI & ML) — *June 2023 – May 2027*
CGPA: 8.21/10
Kattankulathur, Chennai, Tamil Nadu, India

A-Level (Higher Secondary Education) — *Global College International*
A-Level — *June 2021 – May 2023*
Kathmandu, Nepal

Work Experience

Academic & Research Writing Intern — *City Research Center Pvt. Ltd.*
Kathmandu, Nepal — *May 2025 – July 2025*

- Prepared academic reports, assignments, and research-related documents across various topics.
- Conducted research and organized information into clear, structured reports.
- Assisted in preparing academic content and documentation for research and educational projects.

Publications

Dynamics-Guided Cross-Modality Masked State Space Modeling for Robust Wearable Human Activity Recognition *2026*
IEEE 5th World Conference on Applied Intelligence and Computing (AIC)

- Co-authored a research paper on self-supervised learning for wearable human activity recognition.
- Proposed a dynamics-guided masking and cross-modality reconstruction framework using state space models.
- Achieved 91.48% accuracy and 91.55% Macro F1-score on the UCI HAR dataset.
- **DOI:** 10.1109/AIC70896.2026.11717057

Projects

Smart Dustbin — *SRM Institute of Science and Technology*
Arduino, Ultrasonic Sensor, Servo Motor, Embedded C — *2024*

- Designed an automated dustbin that opens its lid when a person approaches.
- Integrated an ultrasonic sensor and servo motor with an Arduino to enable touch-free operation.

Obstacle Avoiding Robot — *SRM Institute of Science and Technology*
Arduino, Ultrasonic Sensor, Servo Motor, Embedded C — *April 2024*

- Developed a robot that detects and avoids obstacles using an ultrasonic sensor.
- Integrated Arduino, a servo motor, and sensors to enable autonomous obstacle avoidance.

Awards & Achievements

Certificate of Merit – Best Project *April 2024*
TECHKNOW 2023–2024, SRMIST

- Presented and demonstrated an Obstacle Avoiding Robot project.
- Recognized as the best project/working model by the expert technical committee.

Teaching & Mentoring Experience

Python Programming Mentor *June 2025*
Shree Ratna Rajya Secondary School *Baneshwor, Kathmandu, Nepal*

- Mentored Grade 8 students in Computer Science for one week.
- Conducted interactive sessions on Python programming and encouraged student participation.
- Explained programming concepts through discussions and practical learning activities.

Technical Skills

Programming Languages	C, C++, Java, JavaScript, Python
Data Visualization & BI Tools	Power BI, Tableau, Microsoft Excel
Database Tools	SQL, SQL Server Management Studio (SSMS 22)
Other Tools	LaTeX

Language Proficiencies

English	Proficient
Nepali	Fluent
Maithili	Fluent
Hindi	Fluent